ONERHYTHM

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PROJECT DE NOVO

Project Brief

Built anew for a problem that is still too large to ignore.

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> *“As a patient for 10 years dealing with high burdens of virtually all types of arrhythmia, it was forced problem solving. Think I might have gotten alarmingly good at it.”*

EXECUTIVE BRIEF

Executive Brief

Project De Novo begins with an uncomfortable technical fact: you are one human, and the problem is huge. If the goal is to contribute meaningfully to the intersection of psychological distress and arrhythmia, then the requirements quickly become larger than any single founder operating manually can reliably hold. You need access to meaningful data. You need GPU time. You need a way to think clearly across clinical complexity, research sequencing, engineering execution, organizational governance, and long-horizon product design. If you can get the data and the compute, you can contribute to solving real problems. But even then, the bottleneck does not disappear. It changes shape. The problem becomes whether one person can create enough structure to make the work compound rather than fragment.

Project De Novo is the answer to that constraint. It is the operating architecture for pursuing a very large problem with one human at the center and a carefully orchestrated set of AI systems around him. The name is intentional. De Novo means built anew. It signals a willingness to start from first principles when the scale of the challenge makes inherited assumptions insufficient. It also reflects the reality that work at the edge of psychocardiology, machine learning, and regulated health-adjacent infrastructure cannot be built casually. Even in the abstract, it is clear that serious work in this domain will meet scrutiny. That means rigor is not a downstream activity. It is an opening condition.

THE TECHNICAL PROBLEM

The Technical Problem

The technical problem is not that arrhythmia-related psychological distress is difficult to model. The technical problem is that the underlying system required to pursue it is itself large, interdisciplinary, and unforgiving. Data access is hard. Useful labeling is hard. Compute is finite. Context is fragile. Scope expands faster than insight. Narrative can get ahead of reality long before evidence is mature enough to support it. All of that happens before a single meaningful model result is produced.

A founder working alone in this environment faces a structural asymmetry. The ambition has to be large, because the problem is large. The execution has to be disciplined, because the domain demands it. But the human bandwidth remains fixed. That creates a need for a different kind of leverage. Not just faster writing. Not just faster coding. Operational leverage. Memory. Context retention. Structured review. Scope isolation. Drift mitigation. Repeatable reporting. The point is not to replace human judgment. The point is to preserve it under scale.

RATIONALE

Why De Novo

To build anew is not to ignore what exists. It is to recognize when existing paths are not enough. Project De Novo takes an abstract view of the work from the outset. The question is not only how to train better systems. The question is how to build an environment in which better systems can be pursued coherently by one person without losing truth along the way.

That requires orchestration of AI rather than casual use of it. It requires isolating scope instead of letting every promising idea become active work. It requires managing context so decisions do not dissolve between sessions. It requires mitigating drift so that language, planning, and technical claims do not outrun what has actually been built. It requires treating governance as infrastructure. In that sense, Project De Novo is not merely a research effort and not merely a software effort. It is an operating discipline for carrying an outsized problem in a way that remains intelligible, auditable, and resilient.

ORIGIN

Patient Origin

There is also a human reason this project exists. As a patient for ten years dealing with high burdens of virtually every kind of arrhythmia, problem solving was not optional. It was forced. It became a daily operating condition. When a body refuses to behave predictably, and when the consequences touch work, family, identity, sleep, fear, and survival, the mind adapts by learning to solve hard problems whether it wants to or not. There is no romance in that. It is repetition under pressure.

The logic behind Project De Novo comes from that decade. After enough years of living inside a system that could destabilize without warning, pattern recognition becomes unusually sharp. Constraint handling becomes instinctive. The tolerance for vague thinking gets very low. In plain language: ten years of arrhythmia made me alarmingly good at forced problem solving. Project De Novo is what happens when that learned behavior is redirected toward a problem larger than one patient story. The same instinct that once had to survive a misfiring heart is now being used to build the structure required to pursue a field-level contribution.

METHODOLOGY

The Approach

The approach is straightforward in concept, even if demanding in execution. One human remains responsible for intent, judgment, and consequence. Around that center sits an orchestrated AI environment designed to expand throughput without surrendering coherence. The system must remember what matters. It must preserve context across sessions. It must separate active priorities from attractive distractions. It must force recurring review. It must detect when scope is drifting, when narrative is inflating, and when workstreams are losing contact with evidence. It must make the next decision visible. It must keep the founder from becoming the only place where institutional memory exists.

In practice, that means building an operating model where AI is used as structured organizational capacity rather than as a novelty layer. Research, writing, planning, technical decomposition, audit logic, task reporting, and synthesis can all be accelerated, but only if each function is given boundaries and if the overall system is governed. Project De Novo therefore treats orchestration itself as a technical asset. The advantage is not merely that more can be done. The advantage is that more can be done without the work collapsing into noise.

STRATEGY

Strategic Advantage

The strategic advantage of Project De Novo is not that it makes a large problem small. It does something more useful. It makes a large problem governable. Most early-stage technical efforts fail not because the founders lack intelligence or conviction, but because the project becomes too complex to hold cleanly. Context leaks. Priorities multiply. Assumptions harden without review. Stakeholders receive language that sounds more mature than the underlying work. A single human becomes both bottleneck and single point of failure.

Project De Novo addresses that directly. By treating memory, review, scope discipline, and drift mitigation as first-class infrastructure, it creates a foundation from which a solo founder can operate more like as the commander of a controlled organization. That signals seriousness for research collaborators. It reduces chaos cost for technical partners. It matters to future funders because it suggests that execution is being built on repeatable systems rather than willpower alone.

DIRECTION

Where This Leads

Where this leads is not a single model, a single paper, or a single build. It leads toward a durable capability. If enough data can be reached, if enough compute can be secured, and if the operating architecture remains disciplined, then one human with the right systems can contribute to a category of problems that has historically required far larger institutions to even approach. The destination is not just prediction.

Project De Novo is both practical and directional. Practically, it is the system that allows one founder to pursue a very large research and product vision without surrendering coherence. Directionally, it is a thesis about how hard problems should be approached when modern AI becomes available to individuals. The future does not belong simply to the teams with the most models. It belongs to the teams that can structure intent, manage evidence, isolate scope, and keep truth intact while moving quickly. Project De Novo is an attempt to build exactly that kind of capability from the ground up.

AUTHOR

Author

Matthew J. Adams is the founder of OneRhythm and author of this brief. Project De Novo reflects his effort to build the operational discipline required for one human to pursue a very large research and product vision without surrendering coherence, rigor, or strategic truth.

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